Presentation 1

- Links to presentation(s) and code(s) on GitHub

Google Slides		Dashboard		Code

- What did you do?

New metric proposal: Self-Service Success Rate
Main visualization: Count of Ending User Type For Customer Calls

1. **Preliminary visualizations with Power BI using the AllCallsData**

I collaborated with my team to create basic Power BI visualizations. This involved some simple data cleaning within Power BI (e.g. converting Called number column to Text data type to resolve import errors). Although not every visualization is of equal importance, I created 5 visualizations total:

- Count of Redirect Reason (bar chart)
- Average Duration by Redirect Reason (bar chart)
- Average Duration by User Type (bar chart)
- Proportion of Answer Indicators By User Type (stacked bar chart)
- Count of Ending User Type For Customer Calls (bar chart)

The first 4 visualizations are exploratory, whereas the last visualization (Count of Ending User Type For Customer Calls) is a step towards visualizing my proposed metric.

2. **EDA and data processing for AllCallsData**

Individually, I spent time investigating how customer calls can be identified using columns in the data. I researched what the Inbound/Outbound trunk, Direction, Call type, and Client type columns represented, and I came to the conclusion that incoming calls from customers would satisfy the following conditions in the earliest leg of the call:

- Inbound trunk = NA
- Outbound trunk = wcc_[something]
- Direction = TERMINATING
- Call type = SIP_INBOUND
- Client type = WXCC

This was used in my data cleaning code in Python. I removed columns that were exclusively NaN, converted all phone number columns to string instead of integer/float, and converted columns with mixed data types to entirely strings. Then, I grouped the data by Correlation ID. Using the conditions I came up with to detect calls that are specifically inbound customer calls, I filtered the Correlation IDs. This yielded a collection of data frames that each represent one customer call.

I also explored the User type column, which can indicate if a call reached a live agent (User type = User), if a leg corresponds to a customer still navigating the submenu (User type = AutomatedAttendantVideo), if a call ended with a voicemail (end leg's User type = VoiceMailRetrieval), etc.

For each possible User type value, I created a small table that contains the counts of how many customer calls have that User type value in the ending leg. This table was fed into Power BI to create the main visual.

3. **New metric: Self-Service Success Rate**

This metric is calculated by how many calls that never involved a live agent did NOT end with a customer abandoning the call. This metric would be important in showing how effectively customers will be able to complete their task without needing a live agent. This can help identify ways to reduce the call load on live agents, streamlining the call process.

In the data, this would involve steps like looking at what calls never have a leg with User type = User. Identifying whether a call ended with a customer abandoning the call could involve using the Original/Redirect reason, Call outcome, Call outcome reason, and Answer Indicator columns.

Mathematically: # of self-service calls that did not end in abandonment / total # of self-service calls.

- How does it help the project?

The preliminary visualizations were to help us get the hang of Power BI as a tool and to understand the data better with hands-on work.

For my main visualization, it helps the project as it depicts the distribution of call outcomes for customer calls specifically. Although there are more columns that detail the exact way in which calls end, the User type column is especially useful for this as it shows info like whether a call ended in voicemail or a live agent. With this kind of info, we can identify points of improvement in the phone system if, say, calls ending in voicemail is the dominant way in which calls may be considered as unsuccessful.

Looking at the visualization, it is visible that the User column (i.e. calls that ended in live agents) is noticeably smaller than the columns for the AutomatedAttendantVideo (i.e. calls that ended with the customer still in the submenu) and for the VoiceMailRetrieval (i.e. calls that ended with voicemail). This suggests that there is a relatively high rate of customers leaving while trying to navigate the submenu, and calls that end with a live agent are rare. This can serve as a starting point for identifying where issues are present in call journeys.

For my metric, it helps the project by allowing us to identify weak points in the submenu/routing system, and the ultimate goal would be to maximize the metric as that would mean less call loads for live agents. My metric would allow us to also gauge how well the company is performing in that regard. If the metric is low (i.e. high rate of abandonment without live agents), then identifying at what point those calls are abandoned could be helpful in determining a more specific point of improvement.

My data cleaning process will also be very helpful going forward as it organizes the data into individual calls that can be analyzed and compared. This will help us in getting a better understanding of how different call journeys are represented differently by the data. It will also make it easier to get info that cannot be easily obtained by simply looking at each row/leg separately.

- Issues faced (if any)

Errors in importing the data to Power BI. Inability to share dashboards further complicated the issue my group had with me being the only one with a Windows PC. Trying the run the grouping code for the full AllCallsData ended up taking over 30 minutes,

- Attempts to resolve issues (if any)

Power BI import errors were resolved by simply converting a column (“CalledNumber”) to strings. The Power BI computer issue with my groupmates all having Macs was resolved by trying to use library computers. We also ended up just making separate dashboards for our individual metrics and put aside the idea of sharing one pbix file for now. For the excessive runtime of my code, I only used the August 2025 data for now.

- Issues resolved (if any)

Power BI import errors were fixed. We managed to make something work with Power BI despite our different computer systems (and we hope to have Power BI Pro access soon). The excessive runtime for my code on the entire AllCalls dataset is still unresolved.

- Next steps

The biggest next step I have is to identify ways to use the AllCalls data to determine whether a call was abandoned or not and whether a call can be considered a success. I have some specific columns in particular to research and understand: Original/Redirect reason, Call outcome, Call outcome reason, and Answer Indicator. These appear to be potentially useful as they all give information about the state of a call at any given leg, and a call that was abruptly ended by the customer could be indicated by these columns. After understanding these columns better, I plan to use them in a conditional statement that can essentially label call sessions as successful or abandoned using Python code.

Another major next step is figuring out ways to rewrite the grouping code that will make splitting the combined AllCalls dataset into separate customer calls take less time to run. I suspect that using a for loop to loop over different dataframes with dozens of columns might be inefficient, so I think somehow reducing the size of the dataframes for the sake of just getting the correlation IDs that correspond to customer calls could help.

- References (Mention if you built up on someone else's work)

n/a